

Score: 22 %

Calculations Quiz

1. Calculate the relative molecular / relative formula mass (M_r) of the following;

a) Aluminium carbonate

$$M_r(\text{Al}_2(\text{CO}_3)_3) = (26.98 \times 2) + (12.01 \times 3) + (16 \times 9) \\ = 233.99$$

(ii)
b) Cobalt chloride pentahydrate

$$M_r(\text{CoCl}_2 \cdot 5\text{H}_2\text{O}) = 58.93 + (35.45 \times 2) + (10 \times 1.008) + (5 \times 16) \\ = 219.91$$

(4 marks)

2. If you had 0.792 moles of sodium oxide powder calculate;

a. The number of Na_2O formula units present.

b. The number of sodium ions present.

a)
$$N(\text{Na}_2\text{O}) = n \times A_v \\ = 0.792 \times 6.022 \times 10^{23} \\ = 4.77 \times 10^{23} \text{ formula units}$$

b)
$$N(\text{Na}^+) = \underline{4.77 \times 10^{23}} \times 2 \\ = 9.54 \times 10^{23} \text{ ions}$$

(3 marks)

1 mark for M_r included in answer
1 mark for correct answer

2 marks for part (a) 1 for working, one for correct answer
1 mark for correct answer part (b)

2 marks for correct calculations
1 mark for answering question

3. If you had the same mass of two copper containing minerals, $\text{Cu}_2\text{CO}_3(\text{OH})_2$ and CuFeS_2 , use percent composition to determine from which you would be able to extract a higher amount of copper.

$$\%(\text{Cu}) = \frac{(63.55 \times 2)}{(63.55 \times 2) + 12.01 + (16 \times 5) + (1.008 \times 2)} \times 100$$
$$\text{Cu}_2(\text{CO}_3(\text{OH}))_2 = 57.5\%$$

$$\%(\text{Cu}) = \frac{63.55}{63.55 + 55.85 + (32.07 \times 2)} \times 100$$
$$\text{CuFeS}_2 = 34.62\%$$

You would extract more copper from $\text{Cu}_2\text{CO}_3(\text{OH})_2$ (3 marks)

4. If you needed 8.44 moles of potassium hydroxide pellets, what mass would you need to weight out?

$$m(\text{KOH}) = n \times M$$
$$= 8.44(39.10 + 16 + 1.008)$$
$$= 473.55\text{g}$$

(2 marks)

1 mark for working
1 mark for answer
-0.5 if no units

5. You have some limestone available to you and chemical analysis has determined that it is 79% pure calcium carbonate. You wish to extract 85 g of elemental calcium from the limestone.

- Calculate the percent composition of calcium in CaCO_3 .
- Calculate the mass of limestone you would require to extract the required amount of calcium.

a) $\%(\text{Ca})_{\text{CaCO}_3} = \frac{40.08}{40.08 + 12.01 + (16 \times 3)} \times 100$
 $= 40.04\%$

b) $m(\text{CaCO}_3) = 85 \div \left(\frac{40.04}{100} \right) = 212.29 \text{ g}$

$m(\text{limestone}) = 212.29 \div \left(\frac{79}{100} \right) = 268.72 \text{ g}$

(3 marks)

6. If you had 1.08 kg of silver oxide, how many silver atoms would be present?

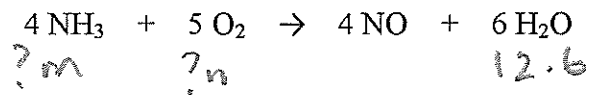
$n(\text{Ag}_2\text{O}) = \frac{m}{M}$
 $= \frac{(1.08 \times 1000)}{(107.9 \times 2) + 16}$
 $= 4.659 \text{ mol}$

$N(\text{Ag}_2\text{O}) = n N_A$
 $= 4.659 (6.022 \times 10^{23})$
 $= 2.8057 \times 10^{24} \text{ formula units}$

(3 marks)

$N(\text{Ag}) = 2.8057 \times 2$
 $= 5.61 \times 10^{24} \text{ silver atoms}$

7. Consider the following reaction.



If 12.6 g of water was produced, calculate;

- The number of moles of oxygen consumed.
- The mass of ammonia consumed.

1 mark a)
$$n(\text{H}_2\text{O}) = \frac{m}{M} = \frac{12.6}{16 + (1.008 \times 2)} = 0.69994 \text{ mol}$$

1 mark
$$n(\text{O}_2) = \frac{0.69994}{6} \times 5 = 0.5828 \text{ mol of O}_2$$

1 mark b)
$$n(\text{NH}_3) = \frac{0.69994}{6} \times 4 = 0.4663 \text{ mol}$$

(4 marks)

1 mark
$$\begin{aligned} m(\text{NH}_3) &= nM \\ &= 0.4663 (14.01 + (1.008 \times 3)) \\ &= 7.94 \text{ g} \end{aligned}$$

-1/2 no units